

AshGridX Technical & Operational Clarification Response

Prepared for BuiltRight Services Ltd | Date: August 2026

1. Executive Summary & Onboarding Procedure

This document provides formal technical responses to the clarification request submitted by BuiltRight Services Ltd. It outlines onboarding steps, pilot setup details, endpoint mappings, webhook verification, and system capabilities.

Pilot Test Procedure & Onboarding Steps

1. Place an order for the purchase of a pilot device.
2. Upon order placement, BuiltRight will be onboarded to the staging environment, enabling API key generation, webhook setup, and IP whitelisting.
3. Installation: Install the pilot device on-site, in close physical proximity to the monitored asset (ensuring minimal/no room for malicious intervention between the asset and load).

Required Onboarding Information: To complete onboarding, BuiltRight should provide:

- Email address of the organization owner
- Full legal name of the organization
- Preferred organization logo

Support Contact Information

Support Name	Israel Oluwole
Email Address	ioluwole@ashipaelectric.com
Phone Number	+2348166473017
Operating Hours	Monday – Sunday, 8:00 AM – 5:00 PM (24/7 coverage for critical exigencies)

2. Core Capabilities & Protection Mode

Protection Mode & Removal: When a bypass or tamper attempt occurs, the AshGridX device automatically enters protection mode and cuts off power output. To restore operations, the `POST /device/remove-protection` endpoint must be called to request a security code, which is then input into the device. Until this process is completed, the device remains in protection mode.

Device Control Execution (204 No Content): A HTTP 204 response from `/device/control` represents successful acceptance and processing of the control request within the business process. The device control workflow is event-driven; state change is eventual rather than instantaneous.

3. Device Onboarding, Assignment, and Identity

Q1: How does BuiltRight register a new device through the API?

Status: **Not currently supported**

Response: APIs are not exposed for device registration/onboarding. Devices are onboarded via dashboard to maintain a simpler user experience.

Q2: Can device registration and customer assignment be automated, or must they be performed in the AshGridX dashboard?

Status: **Supported but documentation pending**

Response: It is advised to perform device onboarding via the AshGridX dashboard. End-to-end APIs may be exposed in future releases.

Q3: Is customerDeviceId generated by AshGridX or supplied by BuiltRight?

Status: **Supported and documented**

Response: customerDeviceId is generated and provided by AshGridX. BuiltRight should use deviceNumber as the deterministic, independent identifier.

Q4: What is the relationship between deviceId, customerDeviceId, deviceNumber, inventoryNumber, siteId, ownerId, and accountNumber?

Status: **Supported and documented**

Response: • customerDeviceId: Internal AshGridX reference.

- deviceNumber: Deterministic, primary customer identifier.
- siteId: AshGridX-generated site/project identifier.
- ownerId: Identifier for assigned customer.
- inventoryNumber: Onboarding step reference leading to operational readiness.
- deviceId: Generated internal ID.
- accountNumber: Optional field.

Q5: The control endpoint uses customerDeviceId, while the webhook example uses deviceId. How should BuiltRight map these values reliably?

Status: **Supported and documented**

Response: deviceId in the prior webhook example was a documentation typo and refers to deviceNumber. BuiltRight should map using deviceNumber and customerDeviceId.

Q6: Is there an endpoint to list all devices belonging to BuiltRight and retrieve their assigned customers/sites?

Status: **Supported but documentation pending**

Response: Supported internally; endpoint can be exposed upon request.

Q7: Is there an endpoint to retrieve one device by customerDeviceId, deviceId, or deviceNumber?

Status: **Planned with expected delivery date**

Response: Supported internally; additional endpoints to retrieve by deviceNumber will be exposed.

Q8: How is a device replaced while preserving customer, project, and control-history relationships?

Status: **Supported but documentation pending**

Response: An internal API handles device replacement. Utilizing customerDeviceId for information retrieval preserves relationships.

Q9: Can one customer or installation have multiple AshGridX devices?

Status: **Supported and documented**

Response: Yes, multiple devices can be linked to a single customer/installation.

Q10: Can a device be unassigned, decommissioned, or reassigned through the API?

Status: **Supported but documentation pending**

Response: Supported internally through standard workflows; specific API endpoints are not currently exposed.

Q11: What fields are mandatory when a customer device is created, and which customer information is stored by AshGridX?

Status: **Planned with expected delivery date**

Response: When the creation API is publicly exposed, full mandatory field specifications and data mechanics will be documented.

4. Remote Control, Status Verification, and Protection

Q1: Does a 204 response from /device/control mean the command was queued, transmitted, acknowledged, or physically executed?

Status: **Supported and documented**

Response: A 204 status confirms successful API request acceptance for the business process. Execution is event-driven.

Q2: How can BuiltRight retrieve the device's live on/off state after issuing a command?

Status: **Supported and documented**

Response: Retrieve device state by querying via customerDeviceId or deviceNumber.

Q3: Does device-control history represent requested state or confirmed physical execution?

Status: **Supported and documented**

Response: Device-control history represents the requested state.

Q4: Is there a command reference or transaction ID that can be used for idempotency, reconciliation, and support?

Status: **Planned with expected delivery date**

Response: Command/transaction ID tracking can be considered for inclusion in future iterations.

Q5: What happens when a device is offline when an on/off command is submitted?

Status: **Supported but documentation pending**

Response: The request is processed and command issued, but not effected while offline. Future iterations will enforce eventual execution upon reconnection.

Q6: Are commands queued for later execution, and if so, for how long?

Status: **Not currently supported**

Response: Command queuing for offline devices is not supported currently.

Q7: Can control commands fail after the API initially returns success? How is that failure reported?

Status: **Supported and documented**

Response: Control requests process synchronously at API level—if failure occurs during request validation/submission, the call fails immediately.

Q8: Is there a webhook for CONTROL_ACCEPTED, CONTROL_EXECUTED, CONTROL_FAILED, or CONTROL_TIMED_OUT?

Status: **Planned with expected delivery date**

Response: Not currently implemented, but webhooks for CONTROL_ACCEPTED, CONTROL_EXECUTED, and CONTROL_FAILED are planned.

Q9: Are repeated requests idempotent, and how should BuiltRight prevent duplicate commands?

Status: **Not currently supported**

Response: Idempotency is not managed server-side; BuiltRight should manage command deduplication on the client application layer.

Q10: Is bulk device control supported or planned?

Status: **Planned with expected delivery date**

Response: Bulk control is currently not supported; can be planned based on operational requirements.

Q11: What exactly does /device/remove-protection do at device and inverter level?

Status: **Supported and documented**

Response: Generates the protection removal key required to restore power output following a tamper/bypass trip.

Q12: Is remove-protection reversible? Is it different from setting control to on?

Status: **Supported and documented**

Response: Not reversible except when the device trips back into protection mode upon a new event. It is distinct from standard on/off control.

Q13: What authorization level is required for remove-protection, and can it be disabled for BuiltRight's API key?

Status: **Supported and documented**

Response: Remove-protection permissions can be granularly restricted or completely disabled for designated API keys.

5. Tamper, Bypass, and Connectivity Alert Requirements

Q1: Does BYPASS mean physical tampering, cable disconnection, communication bypass, meter bypass, protection removal, or another condition?

Status: **Supported and documented**

Response: BYPASS indicates physical tampering or cable disconnection.

Q2: Does unplugging the cable between the AshGridX device and inverter trigger BYPASS immediately?

Status: **Supported and documented**

Response: Yes. Disconnecting the cable from the device terminal immediately trips physical protection on-site.

Q3: Which protected cables, ports, sensors, or enclosure events can generate a tamper alert?

Status: **Supported and documented**

Response: Events originating from the physical device terminal generate tamper alerts.

Q4: What is the maximum expected delay between physical tampering and webhook delivery?

Status: **Supported and documented**

Response: Physical protection trips immediately. Software webhook notifications are dispatched within approximately 5 minutes maximum.

Q5: Does loss of internet or device connectivity generate a separate DEVICE_OFFLINE or DISCONNECTED event?

Status: **Planned with expected delivery date**

Response: Connectivity status monitoring will be remediated in subsequent updates.

Q6: Does reconnection generate TAMPER_RESTORED, DEVICE_ONLINE, or another restoration event?

Status: **Planned with expected delivery date**

Response: Restoration events (DEVICE_ONLINE / TAMPER_RESTORED) are planned for subsequent platform updates.

Q7: Will the payload include deviceId, customerDeviceId, deviceNumber, siteId, and a human-readable tamper reason?

Status: **Supported and documented**

Response: The payload uses the standardized structure with deviceNumber. Human-readable text is omitted as the type field uniquely identifies the event.

Q8: Can BuiltRight query the device immediately after an alert to verify live condition and connectivity?

Status: **Supported and documented**

Response: Yes, immediate status queries are supported.

Q9: How does AshGridX distinguish deliberate tampering from power failure, network outage, maintenance, installation or equipment fault?

Status: **Supported and documented**

Response: Physical tamper detection trips on terminal disconnects. Auxiliary faults (power/network outages) are outside current hardware scope and scheduled for future software remediation.

Q10: Is there a maintenance mode or alert-suppression window for approved technician work?

Status: **Not currently supported**

Response: Maintenance suppression windows are currently not supported.

Q11: Are repeated or continuing tamper conditions re-notified or escalated?

Status: **Supported and documented**

Response: Continuing or repeated tamper conditions generate re-notification webhooks.

Q12: Can historical tamper events be retrieved through an API endpoint?

Status: **Supported and documented**

Response: Yes, accessible via the GET /device/protection-history endpoint.

Q13: Does the physical device retain and later transmit tamper events that occur while connectivity is unavailable?

Status: **Supported and documented**

Response: Yes, tamper events are logged on physical memory and transmitted upon connectivity restoration.

6. Webhook Verification, Delivery, and Security

Signature Payload Structure: The exact structure signed using HMAC SHA-256 Hex is as follows:

```
{
  "event": "value included in the webhook event payload",
  "timestamp": "included in the header (Math.floor(Date.now() / 1000).toString())",
  "data": "the entire payload of the webhook event"
}
```

Q1: What exact byte sequence is signed?

Status: **Supported and documented**

Response: The JSON string representation containing event, timestamp (from header), and data fields is signed.

Q2: Is the timestamp expressed in Unix seconds, Unix milliseconds, or ISO 8601 format?

Status: **Supported and documented**

Response: Unix timestamp in seconds (string format): `Math.floor(Date.now() / 1000).toString()`.

Q3: What timestamp tolerance should BuiltRight apply to prevent replay attacks?

Status: **Supported and documented**

Response: A maximum timestamp tolerance of 2 minutes (120 seconds) should be enforced.

Q4: Must verification use the exact raw HTTP body before JSON parsing?

Status: **Supported and documented**

Response: Yes, raw payload body verification is required.

Q5: How are webhook secrets created, viewed, rotated, and revoked?

Status: **Supported and documented**

Response: Configured via AshGridX Dashboard. Automatic rotation is not currently supported.

Q6: Can old and new secrets overlap safely during rotation?

Status: **Not currently supported**

Response: Overlapping active secrets during rotation are not supported.

Q7: Does every event contain a globally unique eventId for duplicate detection?

Status: **Not currently supported**

Response: Events are dispatched directly without global eventId fields.

Q8: What HTTP status codes does AshGridX treat as successful webhook delivery?

Status: **Supported and documented**

Response: Any 2xx HTTP response status is treated as successful delivery.

Q9: How many times are failed webhooks retried, and what retry schedule is used?

Status: **Supported and documented**

Response: Failed webhook transmissions are retried once currently.

Q10: Are webhook events delivered in order? Can duplicates or delayed events occur?

Status: **Supported and documented**

Response: Events deliver in sequence of occurrence; delays may occur under network constraints.

Q11: Can missed events be replayed from the developer dashboard or retrieved through an API?

Status: **Not currently supported**

Response: Webhook replay functionality is currently not supported.

Q12: What complete webhook event catalogue is supported?

Status: **Supported and documented**

Response: BYPASS is supported; additional network connectivity event types are undergoing active development.

Q13: Can separate webhook URLs be configured for staging and production?

Status: **Supported and documented**

Response: Yes, staging and production dashboards allow separate URL configurations.

Q14: Can webhook sources be restricted by documented IP ranges in addition to signature verification?

Status: **Supported and documented**

Response: IP whitelisting applies strictly to incoming API key requests, not webhook outbound dispatches.

7. API Errors, Limits, Availability, and Support

Q1 & 2: What are the response bodies for validation errors and HTTP error codes?

Status: **Supported and documented**

Response: Errors follow standard Java ResponseStatusException serialized object structures.

Q3 & 4: What request rate limits apply per API key, customer, or device, and are Retry-After headers provided?

Status: **Supported and documented**

Response: Rate limits apply per request with standard rate-limit and Retry-After headers provided.

Q5: What client connection and response timeouts are recommended?

Status: **Supported and documented**

Response: Recommended timeouts: Connection = 10s, Response = 10s.

Q6: What retry policy is safe for GET requests and for device-control POST requests?

Status: **Supported and documented**

Response: Recommended rate limit: 10 RPS with retry-after interval of 5s.

Q7: What uptime or service-level target applies to production control and webhook delivery?

Status: **Supported and documented**

Response: Target service availability is in the 90th percentile.

Q8: Is there a public status page or incident-notification channel?

Status: **Planned with expected delivery date**

Response: A public status page will be made available as soon as possible.

Q9: What logs or correlation identifiers should BuiltRight provide when opening a support case?

Status: **Supported and documented**

Response: Provide relevant application HTTP request/response logs to the dedicated help desk.

Q10: What are the support contacts and escalation hours for a live pilot or device-control incident?

Status: **Supported and documented**

Response: Contact Israel Oluwole (ioluwole@ashipaelectric.com / +2348166473017), Mon-Sun 8 AM - 5 PM (24/7 for urgent escalations).

Q11: Is data encrypted at rest, and what is the retention period for customer/device/control records?

Status: **Supported and documented**

Response: Data is encrypted at rest with indefinite retention currently.

Q12: Where is customer and device data hosted, and what data-protection terms apply?

Status: **Supported and documented**

Response: Hosted in European data centers, encrypted in transit (TLS) and at rest. ISO 27001 and SOC 2 certifications are in progress.